

FalconClean

Interactive Dataset Curator for Duality AI Synthetic Data

TEAM NAME: SynthVision

THEME/TRACK: Duality AI (AI/ML)

Problem Statement

Current Pain Points in Duality AI Workflow

- Manual download, rename, and upload of output.zip takes 10-20 minutes per session.
- Generated images often include low-quality, duplicate, or mislabeled samples.
- Quick visualization of bounding boxes before training is difficult.
- Merging multiple sessions is tedious (duplicate filenames, train/val splits reset numbering).
- Sharing and collaboration on curated datasets is not easy.
- Top performers spend hours on data wrangling instead of modeling.

Result: Severely limits experimentation speed and leaderboard performance.

Proposed Solution

FalconClean - Interactive Dataset Pipeline for Duality AI

A web + notebook hybrid tool to quickly process Falcon sessions:

1. Instant download via Google Storage URL (3-5 sec).
2. Interactive UI for visual image pruning (Accept/Reject with bounding box overlay).
3. Auto-merge unlimited sessions with unique prefixing and proper train/val handling.
4. Visualize statistics (class distribution, bbox sizes, image quality).
5. Export clean, ready-to-train dataset (YOLO format) or upload to Kaggle.
6. Share curated dataset with a single link.

Reduces hours of manual work to under 5 minutes.

Tech Stack

Frontend / Interactive UI: Streamlit (primary) + ipywidgets (Colab backup)

Core Libraries: Python, OpenCV, Albumentations, Pandas, Plotly

Bounding Box Visualization: Custom YOLO drawer (multi-class/color support)

Downloading: requests + Google Storage public URLs

Merging & Renaming Logic: Custom prefix-based system (set_001_, set_002_, etc.)

Deployment Options: Streamlit Community Cloud (free), Google Colab,
HuggingFace Spaces

Optional Future: FastAPI backend + user accounts

Expected Outcome / Impact

Why This Stands Out

- Accelerates dataset creation 10x-20x for Duality/Falcon users.
- Proven to directly boost leaderboard rankings.
- Enables easy collaboration via URL sharing of curated 10k+ image datasets.
- The first visual pruning tool specific to Falcon's output.
- Open-source for quick adoption in Duality Discord & Kaggle.
- Potential for official Duality AI recommendation or platform integration.
- Significant real-world impact on synthetic data quality for robotics and autonomous systems.

Team Details

TEAM LEAD: MUHAMMAD HAARIS

COLLEGE NAME: INSTITUTE OF BUSINESS ADMINISTRATION (IBA) KARACHI

EMAIL: M.HAARISASIF@GMAIL.COM

PHONE: +92 3362888038